

MATH189 Final Report

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Introduction

This project addresses the critical issue of educational equity within the California public school system. Measures of high school success often vary substantially across schools. These differences may reflect disparities in socioeconomic conditions, school resources, and geographic context rather than differences in instructional quality alone.

Understanding how external factors such as income, demographics, and school climate relate to academic outcomes is essential for identifying where achievement gaps are most pronounced. By quantifying the relationship between these factors and school-level outcomes, this project aims to provide insight into structural inequalities that influence educational success and to highlight areas where targeted policy interventions may be most effective.

Prior Research in this Field

Prior research on this topic shows that socioeconomic status is consistently tied to educational outcomes, but the existing literature tends to study either broad academic achievement or individual college attendance rather than school-level college going outcomes like our project focuses on. In Yigiter's study of TIMSS data from 54 countries, the author used integrative data analysis and hierarchical linear modeling to examine both student-level and school-level socioeconomic status (SES). That study found that both individual SES and school SES had significant positive relationships with academic achievement, with school SES showing a stronger effect overall. Yigiter also concludes that these patterns vary across countries and argues that educational policy should address not just individual disadvantage, but also inequalities in school-level resources and opportunities (Yigiter). This research supports many of the same findings and interpretations of the statistical results that we found in our own project, particularly with the correlation between school SES and academic achievement and the need for targeted educational policies to address school-level disparities.

A second particularly relevant paper is Tompsett and Knoester's study of family socioeconomic status and college attendance. Using the Education Longitudinal Study and multilevel modeling, they examined whether students attended college at all, whether they attended a four-year versus two-year institution, and how school context contributed to those outcomes. Their results showed that family SES was positively associated with college attendance, and that school-level socioeconomic composition also mattered. Higher concentrations of low-SES students at a school were associated with lower odds of attending college and lower odds of attending a four-year college rather than a two-year college. They also found that school-level SES remained relevant even when additional factors such as extracurricular activities, expectations,

and test scores were added to the models, supporting the idea that school context helps shape postsecondary outcomes beyond individual background alone (Tompsett and Knoester).

These studies are closely related to our project because they support the same general idea that structural inequality matters at more than one level. Yigiter emphasizes that both individual and school-level socioeconomic conditions are linked to academic success, while Tompsett and Knoester show that family SES and school socioeconomic composition are both associated with college attendance patterns. That aligns with our own emphasis on variables such as FRPM eligibility, absenteeism, stability, and academic readiness as school-level indicators tied to postsecondary outcomes. At the same time, our project fills a gap left by these papers. Yigiter focuses on international math and science achievement at the 4th and 8th grade levels rather than college-going outcomes at the high school level, and Tompsett and Knoester analyze student-level college attendance using national longitudinal data rather than the California school-level administrative dataset that we used. Our project instead centers specifically on College Going Rate across California public high schools, using methods from this course such as exploratory data analysis, hypothesis testing, and multiple linear regression to study how multiple school-level factors relate to postsecondary enrollment. In that sense, our analysis builds on the prior literature while contributing a more localized, school-level view of how socioeconomic and institutional conditions are associated with college-going behavior.

Data Description and Preparation

The primary data source for this project is the [California Department of Education](#) for the 2024-2025 academic year. This recent timeframe was specifically chosen to avoid the highly skewed data from the peak of the COVID-19 pandemic, ensuring a more accurate representation of the current relationships between external factors and performance statistics.

In this project, we analyze the effect of various external factors on our primary dependent variable, College Going Rate. College Going Rate serves as a strong proxy for high school success because it reflects not only graduation but also the degree to which a school equips its students to pursue higher education. The California Department of Education provides extensive supplemental demographic and socioeconomic data, which we utilize to explain the variance in this outcome.

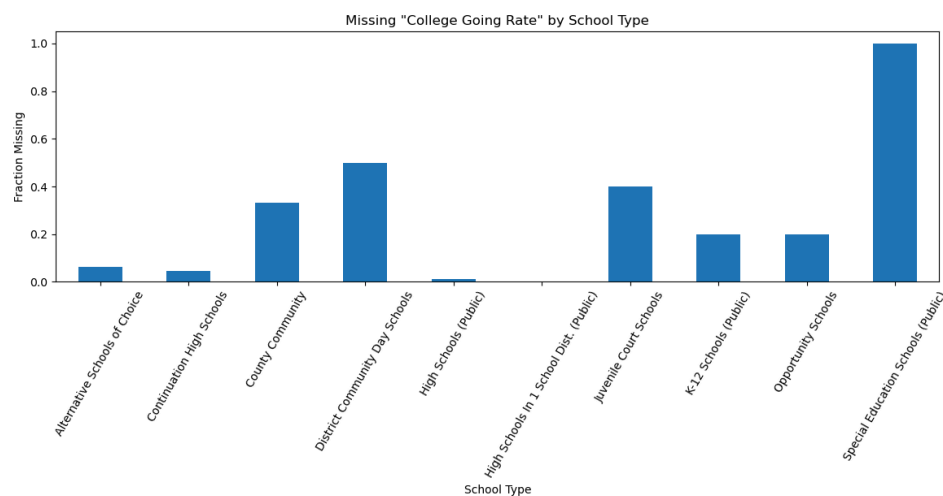
The raw data was provided as separate text files for various categories, including absenteeism, assessment scores, and homelessness. A significant portion of time was dedicated to data collection, parsing, and cleaning to consolidate these disparate sources into a workable format. Since our analysis focuses on school-level performance, we filtered the data to isolate this specific aggregation level. We also discovered and resolved a data quality issue where district offices were labeled with a school code of 0, removing these rows to ensure each high school had a single, unique identifier.

After merging the cleaned dataframes on the school code column, our centralized dataset initially contained 1108 California high schools. The dataset includes categorical descriptors,

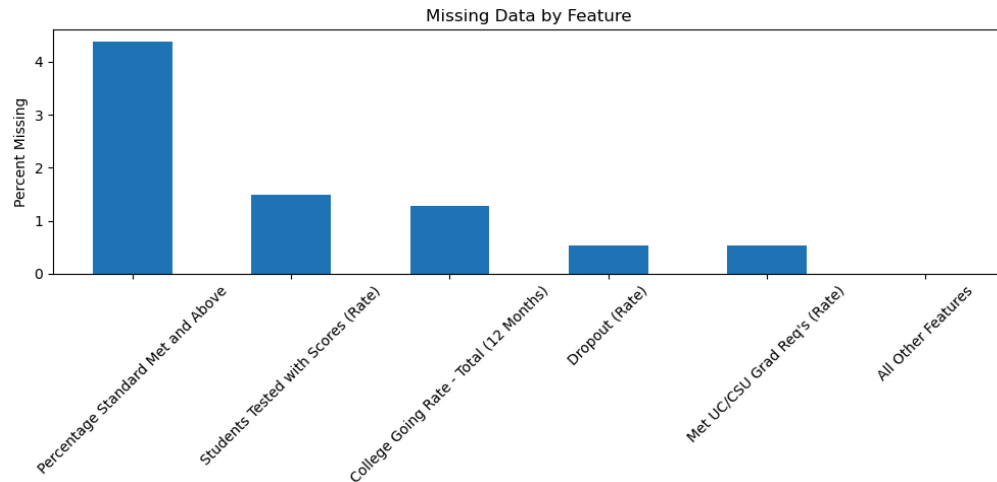
such as charter school status, and numerical indicators like chronic absenteeism rates, suspension rates, and student-teacher ratios.

During our exploratory data analysis, we addressed two primary types of missing data. First, the standardized testing features, specifically Percentage Standard Met and Above and Students Tested with Scores, had minor missingness. Second, we addressed asterisk values scattered throughout the dataset, which the California Department of Education uses to suppress data and protect student privacy for small subgroups. We converted these suppressed values to standard null values for consistent numerical processing.

When analyzing the missingness of our dependent variable, College Going Rate, we observed that typical public high schools had very few missing values compared to other school types.

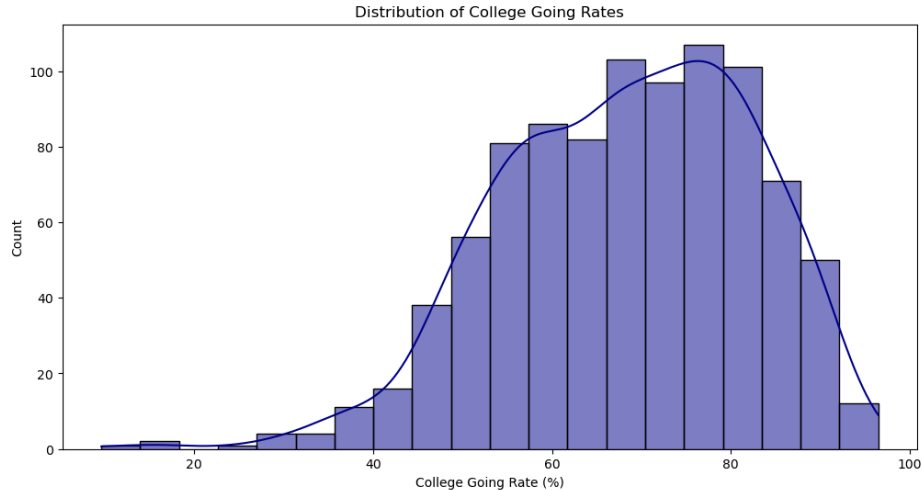


Because non-traditional schools often have different policies or exemptions, and because traditional public schools already accounted for the vast majority of our data, we narrowed our sample to strictly California Public High Schools. Filtering for this specific school type reduced our dataset from 1108 to 935 schools. Of these 935 schools, 923 had valid College Going Rate data. We will address the handling of these remaining missing values in the context of our multiple linear regression analysis. For the regression analysis specifically, we kept only the 890 schools that had valid values across all eleven predictor variables. The graph below illustrates the final missingness across all features in our filtered dataset.



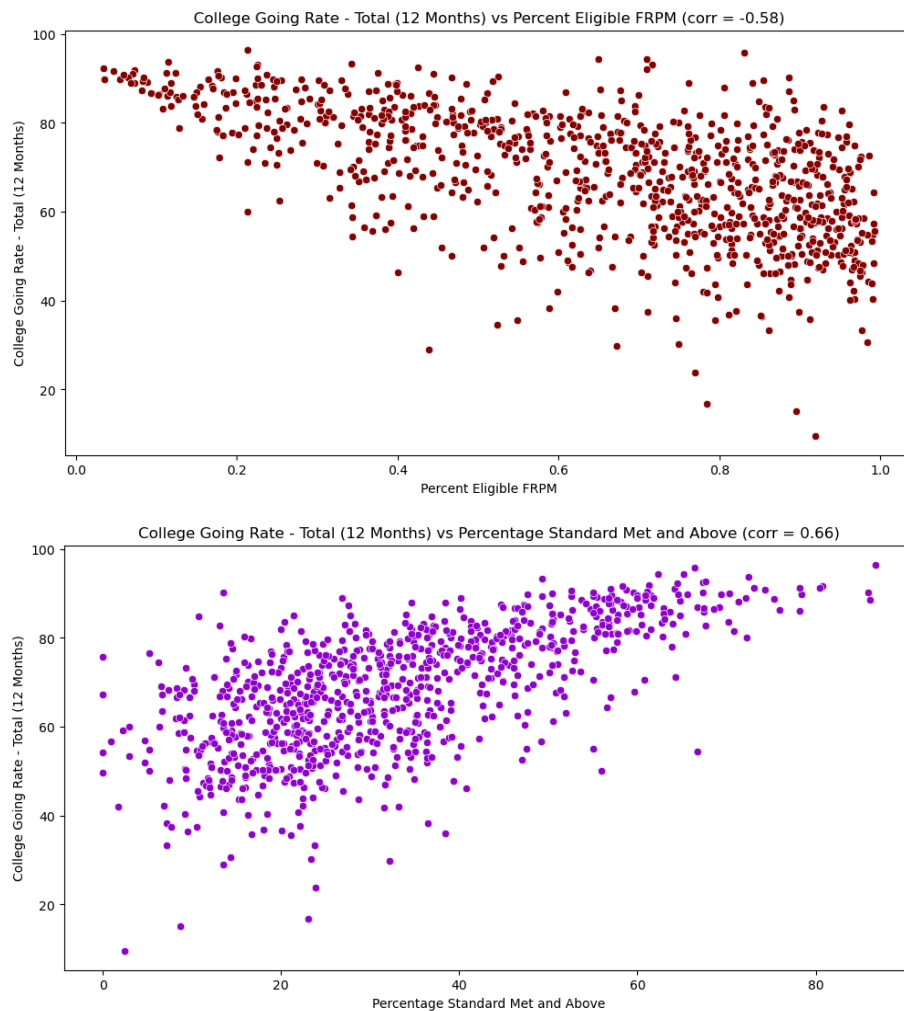
Exploratory Data Analysis

With our dataset finalized to 935 California public high schools, our next step was to understand the distribution of our primary dependent variable, College Going Rate (CGR). The distribution of CGR across these schools reveals a mean of 68.29% and a standard deviation of 13.94%. The data is fairly well distributed across the state, providing a solid baseline for our subsequent modeling.

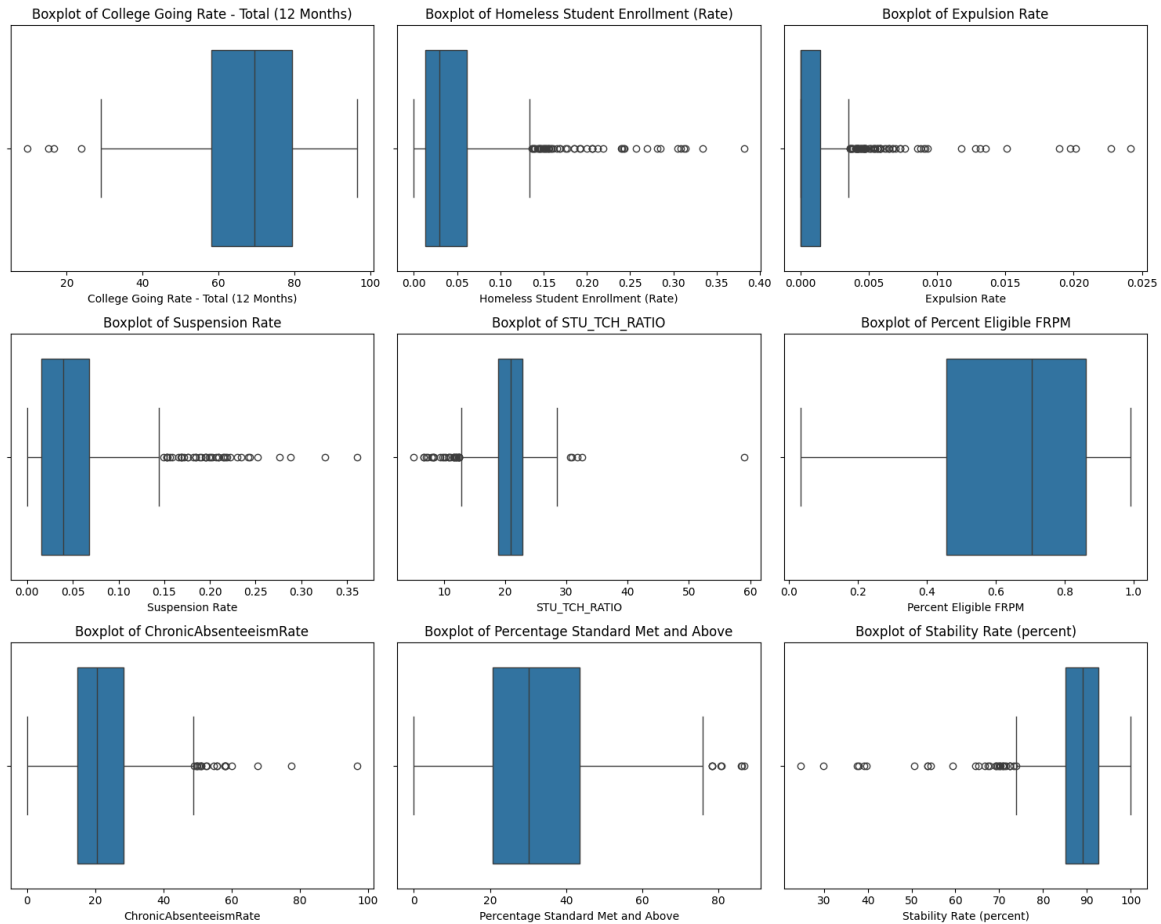


To identify which external factors have the most significant influence on this baseline, we examined bivariate relationships between CGR and our continuous predictors. We specifically looked for the strongest positive and negative correlations to understand what forces drive college enrollment up or down. Academic readiness, measured by the percentage of students meeting or exceeding state standards, showed the strongest positive correlation at 0.656. School stability, measured by the percentage of students enrolled for the entire academic year, also demonstrated a highly positive relationship with a correlation of 0.616. On the other end of

the spectrum, the percentage of students eligible for Free or Reduced-Price Meals (FRPM), a standard indicator of poverty, demonstrated the strongest negative correlation at -0.581.



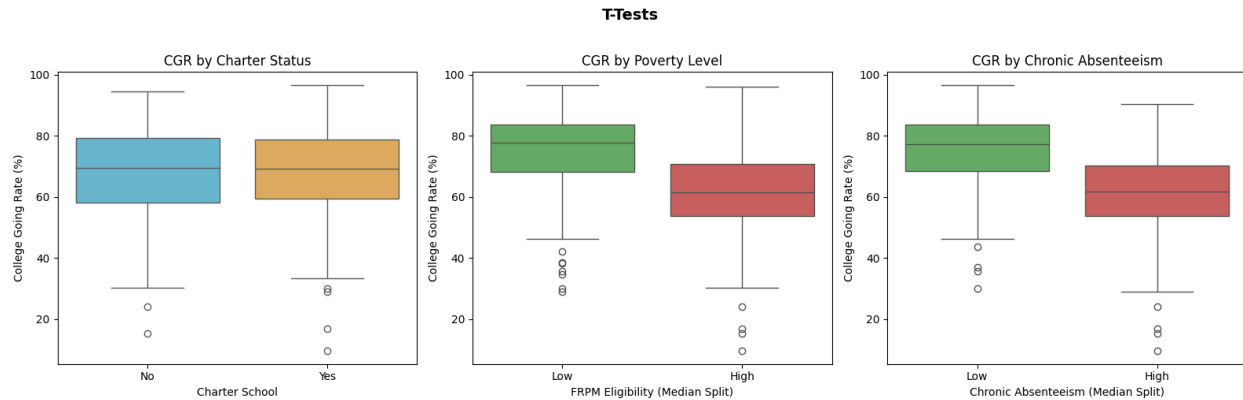
To ensure the robustness of our data before formal modeling, we generated boxplots for all numerical variables to examine their distributions and identify potential outliers. As illustrated in the grid below, while variables such as College Going Rate and Percent Eligible FRPM are relatively well-distributed, features like the Expulsion Rate, Suspension Rate, and Homeless Student Enrollment exhibit heavy right-skewed outliers.



Rather than removing these data points, we chose to retain them. They represent valid, real-world extremes within the California public school system rather than data entry errors. Ultimately, our exploratory analysis of these continuous distributions and their bivariate scatterplots against CGR confirms that both socioeconomic challenges and academic readiness are strongly intertwined with college enrollment. This comprehensive understanding of our dataset's behavior sets a clear, well-founded direction for our statistical testing and multiple linear regression analysis.

Statistical Analyses

To formally test the differences observed in our exploratory data analysis, we first examined the categorical splits for socioeconomic and school climate factors. As illustrated in the side-by-side boxplots below, we evaluated College Going Rate differences based on Charter School status, as well as median splits for FRPM eligibility and Chronic Absenteeism.



We conducted independent-samples t-tests to statistically verify these visual differences. The tests confirmed highly significant disparities when splitting schools by median poverty level ($p < 0.0001$) and median chronic absenteeism rate ($p < 0.0001$), with lower poverty and lower absenteeism groups demonstrating significantly higher mean college-going rates. However, comparing charter schools to traditional non-charter public schools yielded no statistically significant difference in their college-going rates ($p = 0.5356$).

To understand how these continuous and categorical variables interact to collectively explain the variance in college enrollment, we fit a full multiple linear regression model. Our model predicts the College Going Rate using a combination of socioeconomic indicators (such as FRPM eligibility and Homelessness Rate), academic achievement measures (such as Percentage Standard Met), school climate variables (such as Suspension Rate and Chronic Absenteeism), and charter school status.

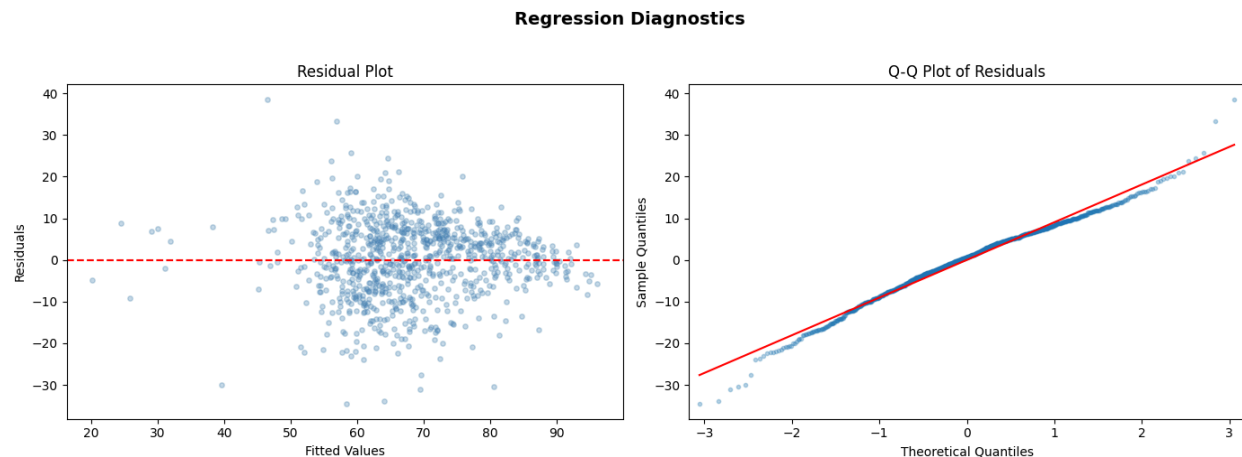
Multiple Linear Regression Results — Predictors of College Going Rate

Predictor	β	SE	t	p -value	95% CI Low	95% CI High
<i>Socioeconomic Indicators</i>						
FRPM Eligibility	-14.9149	1.7792	-8.3831	< 0.001***	-18.4069	-11.4230
Homeless Rate	13.6414	6.9847	1.9530	0.051	-0.0672	27.3500
Enrollment	0.0010	0.0005	2.0546	0.040*	0.0000	0.0019
<i>Academic Achievement</i>						
Pct Standard Met	0.1720	0.0319	5.3959	< 0.001***	0.1095	0.2346
UC/CSU Req Met	0.1478	0.0211	7.0097	< 0.001***	0.1064	0.1892
Dropout Rate	-0.0103	0.0738	-0.1401	0.889	-0.1552	0.1345
<i>School Climate</i>						
Stability Rate	0.4478	0.0626	7.1486	< 0.001***	0.3249	0.5707
Chronic Absenteeism	-0.1334	0.0432	-3.0891	0.002**	-0.2181	-0.0486
Suspension Rate	23.5537	7.6540	3.0773	0.002**	8.5313	38.5760
Stu-Teacher Ratio	0.0089	0.1258	0.0707	0.944	-0.2381	0.2559
<i>School Characteristics</i>						
Charter School (Yes)	-1.6736	1.1615	-1.4409	0.150	-3.9532	0.6061
$N = 890.$ $F(11, 878) = 107.91.$ $R^2 = 0.575.$ Adjusted $R^2 = 0.570.$						

Note. β = unstandardized coefficient; SE = standard error; CI = confidence interval. Significance codes:

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Before interpreting the coefficients, we evaluated the underlying assumptions of our multiple linear regression model. A residual plot was utilized to check for homoscedasticity, showing a relatively random scatter that suggests the constant variance of the residuals was met. Additionally, a Q-Q plot confirmed that the residuals followed an approximately normal distribution.



The overall multiple linear regression model proved to be highly significant ($F = 107.9$, $p < 0.0001$), successfully capturing 57.5 percent of the variation in the College Going Rate across the 890 public high schools that had complete cases for all variables (Adjusted R-squared = 0.5695). Furthermore, an ANOVA nested F-test confirmed that incorporating academic and school climate predictors significantly improved our ability to explain the variance in the College Going Rate beyond what socioeconomic factors alone could capture ($p < 0.0001$).

ANOVA F-Test

- H_0 : All β coefficients = 0 (none of the predictors explain variation in CGR)
- H_1 : At least one predictor significantly explains variation in CGR

To evaluate the explanatory power of our regression model, we conducted an overall ANOVA F-test with the null hypothesis that all regression coefficients are simultaneously equal to zero, meaning none of the predictors explain variation in the College Going Rate. The test returned $F(11, 878) = 107.91$ with a p-value below 0.0001, providing much evidence to reject the null hypothesis. Therefore, we conclude that the model explains a statistically significant portion of the variance in college enrollment across California public high schools.

Nested Model Comparison

While the overall F-test confirms that our predictors collectively matter, we would still like to know if the academic achievement and school climate variables add meaningful explanatory power beyond what socioeconomic factors alone can capture. To address this, we compared a reduced model containing only socioeconomic indicators (FRPM eligibility, Homeless Rate, and

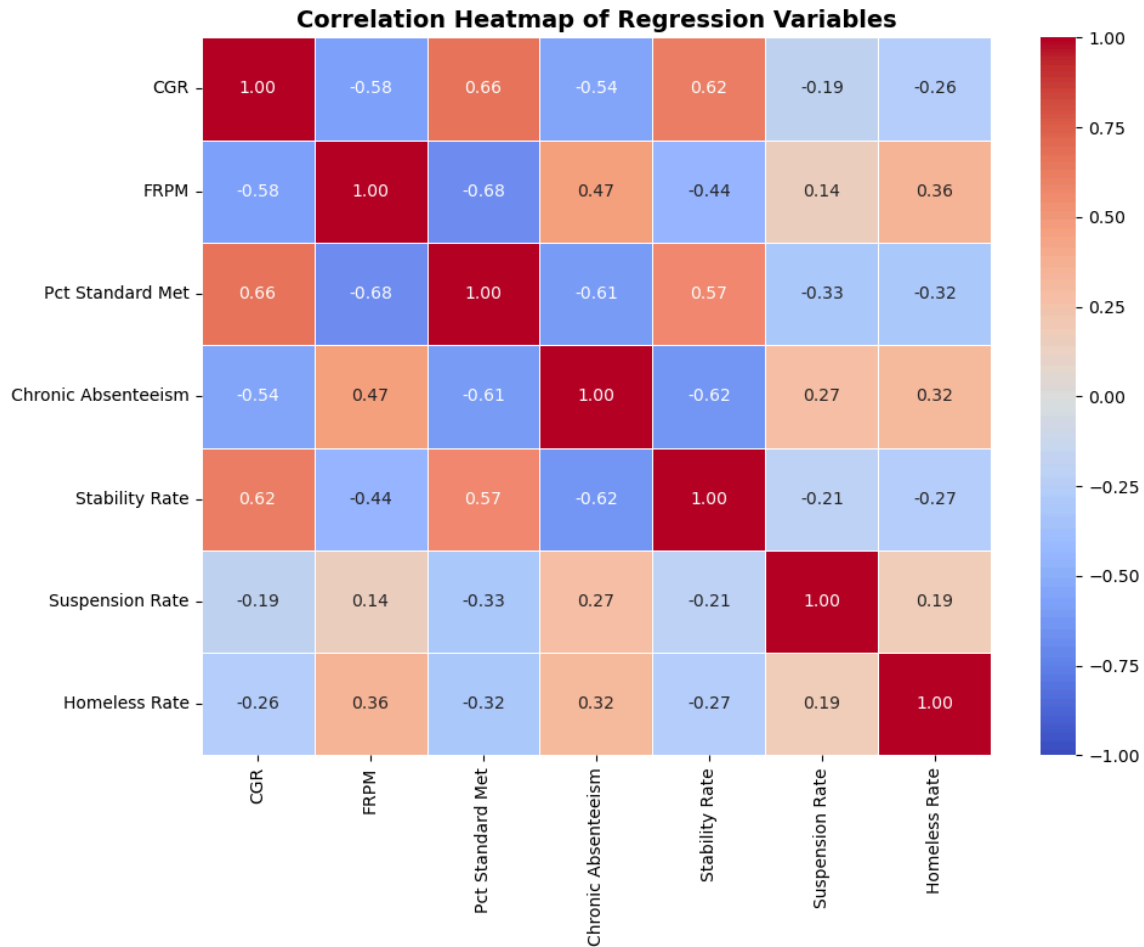
Enrollment) against the full model of all eleven predictors. The reduced socioeconomic model achieved an R-squared of 0.364. This means that poverty and homelessness alone account for around 36 percent of the variance in college going rates. The full model raised this to 0.575, an improvement of 0.210 in explained variance. A nested ANOVA F-test confirmed that this improvement is highly significant, $F(8, 878) = 54.30$ with $p < 0.0001$. This result tells us that academic readiness and school climate variables are not redundant with poverty measures. They provide independent and explanatory power, and add to the view that both socioeconomic barriers and school-level factors must be considered together to understand what drives college enrollment.

Interpretation of Results

The results of our multiple linear regression model reveal several key insights into the factors driving the College Going Rate. Unsurprisingly, academic readiness acts as a strong positive predictor. The model indicates that as the percentage of students meeting or exceeding state standards increases, the college going rate also increases significantly, holding all other variables constant. Similarly, the stability rate shows a strong positive effect, indicating that schools with higher continuous enrollment throughout the year foster better post secondary outcomes.

Conversely, socioeconomic barriers heavily depress college enrollment. The percentage of students eligible for Free or Reduced Price Meals, our primary indicator of poverty, is a massive negative driver in the model. As the percentage of students eligible for these meals increases, the predicted college going rate drops sharply. Chronic absenteeism also serves as a significant negative predictor, further compounding the educational challenges faced by these schools.

When interpreting the coefficients for suspension rates and homeless student enrollment, we must address how they interact with our stronger predictors. In our exploratory data analysis, both of these variables demonstrated a clear negative raw correlation with the College Going Rate. Yet, in the full multiple linear regression model, they paradoxically exhibit positive coefficients.



As illustrated in the correlation heatmap above, suspension rates and homelessness rates share overlapping, moderate correlations with our primary negative predictors, such as FRPM (0.14 and 0.36, respectively) and Chronic Absenteeism (0.27 and 0.32, respectively). Because the meal eligibility and absenteeism variables are so overwhelmingly dominant, they absorb the vast majority of the negative impact associated with systemic educational barriers. Once the model accounts for those primary factors, the remaining, unexplained variance captured by the suspension and homeless metrics becomes mathematically distorted, acting as a structural adjustment rather than a direct negative force. This sign flipping highlights that these factors cannot be evaluated in isolation.

Limitations and Conclusion

While our multiple linear regression model successfully explains a significant portion of the variance in College Going Rates across California public high schools, there are several notable limitations to this analysis. First, due to severe missingness in our dependent variable, we had to exclude non-traditional schools, such as continuation and special education programs. Therefore, our findings are strictly generalizable to traditional public high schools and do not reflect the realities of alternative education pathways. Second, our dataset consists of

aggregated school-level metrics rather than individual student data. This means our conclusions apply to broader school-wide trends and cannot be used to make predictions about individual student outcomes. Finally, because this is an observational study, we can only establish strong statistical associations rather than direct causation. Future research could incorporate unmeasured variables, such as localized district funding allocations, average teacher experience, or granular neighborhood economic data, to capture the remaining unexplained variance in college enrollment.

In conclusion, this project highlights the deep structural inequalities that influence educational success within the California public school system. Our analysis demonstrates that a school's College Going Rate is heavily dictated by overlapping socioeconomic barriers, particularly the proportion of students living in poverty and experiencing chronic absenteeism. While academic readiness and school stability serve as strong positive drivers for post-secondary enrollment, they operate entirely within the constraints of these broader community challenges.

These findings suggest that to effectively close the achievement gap, educational policy interventions cannot focus solely on instructional quality or standardized testing improvements. They must be coupled with systemic community support systems that address fundamental socioeconomic needs, ensuring that all schools have an equitable foundation to propel their students into higher education.

Works Cited

Yigiter, Mahmut Sami. "The Effect of Socioeconomic Status on Academic Achievement: A Big Data Study across Countries and Time with Integrative Data Analysis." *PLOS One*, vol. 20, no. 10, 2025, e0335485. PMC, doi:10.1371/journal.pone.0335485.

Tompsett, James, and Chris Knoester. "Family Socioeconomic Status and College Attendance: A Consideration of Individual-Level and School-Level Pathways." *PLOS One*, vol. 18, no. 4, 2023, e0284188. PMC, doi:10.1371/journal.pone.0284188.